Big Data-Based Social Media Sentiment Analytics Platform

A CAPSTONE PROJECT REPORT

CSA1522-Cloud Computing and Big Data Analytics

to the award of the degree of

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"Big Data-Based Social Media Sentiment Analytics Platform"** has been carried out by **G.Hima Bindu(192511377)** under the supervision of **Dr. P. Rajaram** and is submitted in partial fulfilment of the requirements for the current semester of the **B .Tech- Computer Science and Engineering(AI&ML)** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, G.Hima Bindu (192511377) of the B.Tech- Computer Science and Engineering(AI&ML),SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Diabetes Risk Prediction Application" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
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Total					100%

ABSTRACT

The Cloud-Based Visualization Dashboard is an important module of the proposed Big Data-Based Social Media Sentiment Analytics Platform. It is responsible for presenting the results obtained from social media data collection, processing, and sentiment analysis in a clear and understandable format.

The dashboard displays important information such as positive, negative, and neutral sentiment, emotion levels, topic-wise sentiment, frequently discussed topics, and sentiment trends over time. The processed results from the big data analytics layer are converted into visual representations such as bar charts, pie charts, line graphs, tables, and KPI indicators.

The cloud-based dashboard allows users to access sentiment information from different locations and supports large volumes of continuously generated social media data. Interactive features such as filtering by date, topic, sentiment category, and data source help users perform detailed analysis.

The dashboard can also support real-time or near-real-time monitoring of social media sentiment. If a sudden increase in negative sentiment is detected for a particular topic or product, the system can highlight the change and provide an alert. This helps organizations identify public concerns and respond quickly.

The expected outcome of Module 3 is an interactive, scalable, and user-friendly visualization dashboard that helps users understand large-scale social media sentiment patterns, identify emerging trends, compare opinions, and support data-driven decision-making.

Keywords: Big Data, Social Media Analytics, Sentiment Analysis, Cloud Computing, Data Visualization, Dashboard, Real-Time Analytics

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
BD	Big Data	1
SA	Sentiment Analysis	2

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Social media platforms generate huge amounts of data through posts, comments, reviews, hashtags, likes, shares, and other user interactions. This data contains valuable information about public opinion, customer satisfaction, product perception, brand reputation, and emerging trends. However, the large volume, velocity, and unstructured nature of social media data make conventional methods of analysis difficult and time-consuming.

The proposed Big Data-Based Social Media Sentiment Analytics Platform uses big data and cloud computing technologies to collect, process, analyze, and visualize large-scale social media data. Sentiment analysis is applied to classify social media content into positive, negative, and neutral categories and to identify meaningful patterns and trends.

The complete project is divided into three major modules:

- **Module 1 – Social Media Data Collection Module**
- **Module 2 – Sentiment Analysis & Big Data Processing Engine**
- **Module 3 – Cloud-Based Visualization Dashboard**

Module 1 focuses on collecting and preprocessing social media data. Module 2 performs large-scale data processing and sentiment analysis. Module 3 focuses on presenting the processed analytics results through an interactive cloud-based visualization dashboard.

1.2 Need for the Project

The increasing use of social media has resulted in a continuous generation of user-generated data. Organizations require efficient techniques to understand this information and identify changes in public opinion. A cloud-based visualization dashboard can transform complex analytical results into simple and understandable visual information.

The project is required because:

1. Social media generates large volumes of data continuously.
2. Manual analysis of large datasets is difficult and time-consuming.
3. Organizations need to understand customer and public opinions.
4. Big data technologies are required to process large-scale datasets efficiently.
5. Automated sentiment analysis provides faster and more consistent results.
6. Sentiment trends can help identify emerging issues and public reactions.

7. Cloud-based dashboards provide centralized access to analytical results.
8. Graphical visualization makes complex data easier to interpret.
9. Interactive filtering allows users to analyze specific topics and time periods.
10. Real-time or near-real-time monitoring can support faster decision-making.

1.3 Problem Statement

To design and develop a Big Data-Based Social Media Sentiment Analytics Platform that collects and processes large-scale social media data, performs sentiment analysis using appropriate big data and analytical techniques, and presents meaningful sentiment insights through a scalable and interactive cloud-based visualization dashboard.

1.4 Project Aim

The aim of the project is to develop a scalable cloud-based visualization system that converts processed social media sentiment data into meaningful graphical insights, enabling users to monitor public opinion, identify sentiment trends, compare analytical results, and support data-driven decision-making.

1.5 Project Objectives

The main objectives are:

1. To collect social media data from relevant sources.
2. To clean and preprocess the collected data.
3. To process large-scale social media data using big data technologies.
4. To classify social media content into positive, negative, and neutral sentiments.
5. To identify important topics, keywords, and sentiment trends.
6. To store processed sentiment analytics data efficiently.
7. To design and develop a cloud-based visualization dashboard.
8. To display sentiment results using charts, graphs, tables, and statistical indicators.
9. To provide interactive filtering based on date, topic, source, and sentiment category.
10. To support real-time or near-real-time sentiment monitoring.
11. To identify changes in public opinion through visual trend analysis.
12. To provide meaningful insights for data-driven decision-making.

1.6 Scope

The project covers social media data collection, preprocessing, big data processing, sentiment classification, trend identification, analytics storage, and cloud-based visualization.

The Module 3 – Cloud-Based Visualization Dashboard specifically focuses on presenting processed sentiment information through interactive visualizations. The dashboard can display:

- Positive, negative, and neutral sentiment percentages
- Sentiment distribution charts
- Sentiment trends over time
- Topic-wise sentiment comparison
- Frequently occurring keywords
- Social media source-wise analysis
- Sentiment statistics using tables and indicators
- Interactive filters for date, topic, and sentiment category

The cloud-based approach enables users to access the dashboard through internet-connected devices and supports scalable visualization of continuously generated social media analytics data.

The project does not include development or management of social media platforms, collection of unnecessary private user information, or direct modification of social media content.

1.7 Expected Engineering Outcome

The expected outcome is a functional, scalable, and interactive cloud-based visualization dashboard capable of converting large-scale social media sentiment analytics into meaningful visual insights.

The workflow is:

Social Media Sources → Data Collection → Preprocessing → Big Data Processing → Sentiment Analysis → Analytics Results → Cloud Storage → Cloud-Based Visualization Dashboard → User Insights → Decision-Making

The system is expected to improve the efficiency of social media analysis, provide interactive visualization of sentiment results, identify important sentiment trends, support scalable analytics, and enable faster and more effective data-driven decision-making

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research on cloud-based data visualization, interactive dashboards, sentiment visualization, real-time monitoring, and scalable dashboard systems was reviewed to understand effective methods for presenting the results generated by Module 2 – Sentiment Analysis & Big Data Processing Engine.

The review considered:

- Cloud-based data visualization.
- Interactive analytical dashboards.
- Sentiment result visualization.
- Real-time or near-real-time monitoring.
- Topic-wise sentiment analysis.
- Time-based sentiment trend visualization.
- Interactive filtering and drill-down.
- Scalable charts and graphs.
- Cloud-based dashboard accessibility.
- Data-driven decision support.

The reviewed studies show that interactive cloud dashboards are effective for converting complex analytical results into simple and meaningful visual information. They help users identify sentiment patterns, compare topics, monitor changes over time, and make faster decisions.

2.2 Review of Existing Work

Study 1 – Cloud-Based Data Visualization

Research has investigated cloud platforms for displaying analytical results using charts, graphs, and tables.

Finding: Cloud visualization provides centralized and scalable access to analytical information.

Limitation: Dashboard performance depends on data processing and cloud infrastructure.

Study 2 – Sentiment Visualization

Research has examined methods for presenting positive, negative, and neutral sentiment results using visual charts.

Finding: Sentiment visualization makes public opinion easier to understand.

Limitation: Basic visualization may not provide detailed topic-wise or time-based analysis.

Study 3 – Interactive Dashboards

Research has explored interactive dashboards that allow users to filter and explore analytical data.

Finding: Interactive filters improve data exploration and user understanding.

Limitation: Poor dashboard design can make complex information difficult to interpret.

Study 4 – Real-Time Monitoring Dashboard

Research has focused on dashboards that continuously display changing analytical results.

Finding: Real-time monitoring helps users identify emerging sentiment trends quickly.

Limitation: Continuous updates require efficient data processing and visualization mechanisms.

Study 5 – Sentiment Trend Analysis

Research has investigated graphical methods for representing sentiment changes over different time periods.

Finding: Trend graphs help users identify increases and decreases in public sentiment.

Limitation: Large volumes of time-series data require scalable visualization techniques.

2.3 Comparative Analysis

Table 2.1 – Comparative Analysis of Existing Approaches

Existing Approach	**Advantage**	**Limitation**	**Relevance**
Static Visualization	Simple and easy to implement	Cannot effectively show changing data	Background
Sentiment Charts	Clearly displays sentiment categories	Limited interaction	Suitable
Cloud-Based Dashboard	Scalable and accessible	Depends on cloud infrastructure	Important
Real-Time	Shows updated	Requires	Core

Dashboard	information	continuous processing	requirement
Interactive Dashboard	Supports filtering and exploration	Requires careful interface design	Important
Trend Visualization	Shows sentiment changes over time	Large datasets may become complex	Important
Proposed Cloud Dashboard	Combines sentiment results, cloud access, interaction, and visualization	Depends on reliable processed data	Selected approach

2.4 Research / Engineering Gap

Existing approaches demonstrate the importance of dashboards for presenting analytical information. However, there is a need for a cloud-based interactive dashboard specifically designed to visualize large-scale social media sentiment results generated by the previous processing module.

A dashboard that provides only static charts may not allow users to explore sentiment according to topics, dates, or categories. Similarly, delayed visualization can reduce the usefulness of real-time sentiment monitoring.

Therefore, Module 3 – Cloud-Based Visualization Dashboard addresses this engineering gap by integrating processed sentiment results with interactive charts, graphs, tables, filters, and trend analysis. The module provides a user-friendly interface through which complex Big Data analytics can be converted into understandable visual information.

2.5 Proposed Contribution

The proposed contribution of Module 3 – Cloud-Based Visualization Dashboard is to:

- Display positive, negative, and neutral sentiment.
- Present sentiment trends over time.
- Provide topic-wise sentiment comparison.
- Display frequently occurring keywords.
- Provide interactive date and topic filters.
- Display charts, graphs, and tables.
- Support real-time or near-real-time monitoring.

- Provide centralized cloud-based access.
- Enable easy exploration of analytical results.
- Support faster data-driven decision-making.

Module 3 acts as the visualization and user-interaction layer of the proposed system. It receives processed results from Module 2 – Sentiment Analysis & Big Data Processing Engine and converts them into meaningful visual dashboards for the end user.

Module Flow

Module 1: Social Media Data Collection Module

↓

Module 2: Sentiment Analysis & Big Data Processing Engine

↓

Module 3: Cloud-Based Visualization Dashboard

↓

Charts + Graphs + Tables + Filters + Sentiment Trends

↓

End User / Decision Maker

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholder / User Requirements

Table 3.1 – Stakeholder / User Requirements

Stakeholder	Requirement
Dashboard Administrator	Simple and easy-to-use dashboard
Dashboard Administrator	View overall sentiment results
Dashboard Administrator	Monitor sentiment trends
Dashboard Administrator	Filter data by date and topic
Data Analyst	Topic-wise sentiment comparison
Data Analyst	Access positive, negative, and neutral sentiment results
Data Analyst	View important keywords and topics
Data Analyst	Analyze sentiment changes over time
Developer	Structured visualization data
Developer	Easy integration with sentiment analysis module
System	Reliable presentation of processed sentiment results
System	Support continuous or near-real-time updates
End User	Easy interpretation of charts and graphs
End User	Interactive exploration of sentiment information
Decision Maker	Quick identification of public opinion trends
Application	Support data-driven decision-making

3.1.1 Functional & Non-Functional Requirements

Table 3.2 – Functional & Non-Functional Requirements

Requirement	**Type**	**Target**
Display sentiment results	Functional	Positive, negative, and neutral sentiment
Display sentiment distribution	Functional	Clear percentage/count visualization
Display sentiment trends	Functional	Time-based sentiment trends
Topic-wise sentiment analysis	Functional	Compare sentiment across topics
Display keywords	Functional	Frequently occurring keywords
Filter by date	Functional	User-selected time period
Filter by topic	Functional	Selected topics/categories
Filter by sentiment	Functional	Positive, negative, neutral
Display charts and graphs	Functional	Clear visual representation
Display analytical tables	Functional	Detailed sentiment information
Support dashboard interaction	Functional	Interactive exploration
Update analytical results	Functional	Real-time or near-real-time updates
Cloud-based access	Functional	Centralized dashboard access
Usability	Non-functional	Simple and understandable interface
Performance	Non-functional	Fast dashboard response
Scalability	Non-functional	Supports increasing data volume
Reliability	Non-functional	Consistent visualization
Security	Non-	Authorized dashboard access

	functional	
Availability	Non-functional	Dashboard accessible when required

3.2 Constraints & Applicable Standards

Technical Constraints

- The dashboard depends on processed sentiment data from Module 2 – Sentiment Analysis & Big Data Processing Engine.
- Visualization accuracy depends on the quality of the processed data.
- Large-scale social media datasets may require efficient data processing and visualization.
- Continuous updates may increase computational and storage requirements.
- Dashboard performance depends on cloud infrastructure and network connectivity.
- Data formats must remain compatible between the sentiment analysis engine and visualization dashboard.
- Complex visualizations may require additional processing resources.

Operational Constraints

- Dashboard information should be updated within an appropriate time interval.
- The system should remain usable when large amounts of sentiment data are processed.
- Temporary network problems may affect dashboard accessibility.
- Filters and interactive components should provide results without excessive delay.

Security Constraints

- Dashboard access should be restricted to authorized users.
- User authentication should be implemented where required.
- Sensitive social media information should be handled responsibly.
- Communication between dashboard and cloud services should be secured.
- API credentials and cloud access keys must be protected.

Applicable Standards

The system can follow relevant practices and standards including:

- NIST Cloud Computing Reference Architecture for cloud-based system design.
- ISO/IEC 27001 principles for information security.

- ISO/IEC 27017 guidance for cloud security.
- Appropriate cloud-provider authentication and access-control practices.
- Data privacy and responsible data-handling practices for social media analytics.

3.3 Design Specifications

Table 3.3 – Design Specifications

Parameter	**Requirement**	**Priority**
Sentiment display	Positive, negative, and neutral results	High
Sentiment distribution	Percentage/count of sentiment categories	High
Sentiment trend	Time-based trend visualization	High
Topic analysis	Topic-wise sentiment comparison	High
Keyword display	Important and frequently occurring keywords	Medium
Date filtering	Selectable date/time range	High
Topic filtering	Selectable topics	High
Sentiment filtering	Filter by sentiment category	Medium
Visualization	Charts, graphs, and tables	High
Dashboard interface	Simple and understandable	High
Real-time updates	Real-time or near-real-time results	High
Cloud access	Centralized online access	High
Scalability	Support increasing social media data	Medium
Performance	Fast response and visualization	High
Data security	Authorized access	High
Module integration	Compatible with Module 2 output	High

3.4 Alternative Solutions & Evaluation

Alternative 1 – Static Reports

The system generates static charts and reports from processed sentiment data.

Advantage: Simple to implement and easy to generate.

Limitation: Users cannot interact with the data or easily monitor changing sentiment.

Alternative 2 – Periodic Dashboard

The dashboard updates sentiment results at predefined time intervals.

Advantage: Reduces continuous processing requirements and is relatively simple.

Limitation: Changes in sentiment between updates may not be immediately visible.

Alternative 3 – Interactive Cloud-Based Dashboard

The system continuously or periodically receives processed sentiment results and presents them through interactive charts, graphs, tables, filters, and trend visualizations.

Advantage: Provides better exploration, centralized access, and timely sentiment monitoring.

Limitation: Requires cloud infrastructure and efficient data-processing mechanisms.

Comparative Evaluation

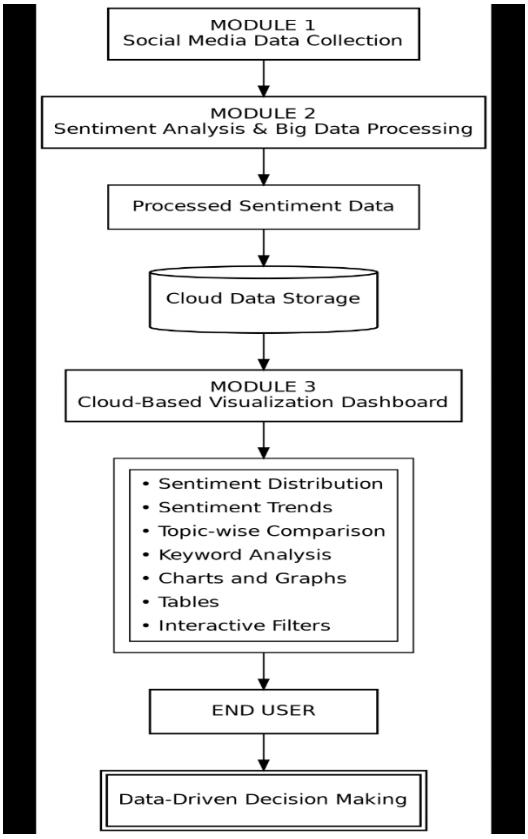
Criterion	**Static Reports**	**Periodic Dashboard**	**Interactive Cloud Dashboard**
Simplicity	High	High	Medium
Interactivity	Low	Medium	High
Real-time visibility	Low	Medium	High
Sentiment exploration	Low	Medium	High
Scalability	Medium	High	High
Topic filtering	Low	Medium	High
Trend analysis	Medium	High	High
User experience	Medium	High	High
Cloud integration	Low	High	High
Decision support	Medium	High	High
Final Selection	—	—	Selected

3.5 Selected Design & Detailed Design

The Interactive Cloud-Based Dashboard approach is selected because it provides an effective way to convert large-scale social media sentiment results into understandable and interactive visual information.

System Workflow

Fig No: 4.1 A Cloud Based Sentiment Analysis and Visualization System for Social Media Data



Data-Driven Decision Making

System Interface

The dashboard should provide clearly visible information for:

- Overall sentiment distribution.
- Positive sentiment percentage.
- Negative sentiment percentage.
- Neutral sentiment percentage.
- Sentiment trends over time.
- Topic-wise sentiment comparison.
- Important keywords.
- Social media data summaries.
- Interactive filters.
- Charts, graphs, and tables.

The user should be able to select a topic, date range, or sentiment category and immediately view the corresponding analytical results.

3.6 Trade-off Analysis

Table 3.4 – Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Interactive dashboard	Static report	Better data exploration	Requires interface development	Selected
Cloud deployment	Local dashboard	Centralized and scalable access	Depends on network	Selected
Real-time updates	Manual updates	Timely information	Requires continuous processing	Selected
Multiple visualizations	Single chart	Provides comprehensive analysis	More design effort	Selected
Interactive filtering	No filtering	Easy exploration of specific data	Requires additional logic	Selected
Topic-wise analysis	Overall sentiment only	Provides detailed insights	Requires topic information	Selected
Trend analysis	Current result only	Shows changes over time	Requires time-series processing	Selected
Modular architecture	Single integrated system	Easier maintenance and expansion	Requires module interfaces	Selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The cloud-based dashboard can support efficient use of computing resources by presenting processed analytical results rather than repeatedly processing the complete social media dataset. Scalable cloud resources can be adjusted according to data-processing requirements.

Ethics

The system should analyse social media data responsibly. User privacy should be respected, and the system should avoid unnecessary collection or exposure of personal information. Sentiment results should be presented objectively without intentionally manipulating public-opinion information.

Health & Safety

The proposed system is a software-based analytics platform and does not directly control physical equipment. However, incorrect analytical results could influence business or organizational decisions. Therefore, visualization and analytical outputs should be validated before being used for important decisions.

Security

The dashboard should:

- Restrict access to authorized users.
- Use suitable authentication mechanisms.
- Protect cloud credentials and API keys.
- Secure communication between system components.
- Protect stored analytical data.
- Maintain appropriate access logs.
- Avoid unnecessary exposure of personal social media information.

Risk Register

Table 3.5 – Risk Register

Risk	**Likelihood**	**Severity**	**Risk Level**	**Mitigation**
Incorrect sentiment data	Medium	High	High	Validate Module 2 results
Delayed dashboard updates	Medium	Medium	Medium	Optimize data update process
Dashboard service failure	Medium	High	High	Use monitoring and recovery mechanisms
Large data volume	High	High	High	Use scalable cloud resources
Slow dashboard response	Medium	Medium	Medium	Optimize queries and visualizations
Incorrect visualization	Medium	High	High	Validate charts and calculations
Network connectivity failure	Medium	Medium	Medium	Use retry and recovery mechanisms
Unauthorized dashboard access	Low	High	Medium	Apply authentication and authorization
Exposure of sensitive data	Low	High	Medium	Apply privacy and access controls
Loss of	Low	High	Medium	Use backup

analytical data				and recovery mechanisms
Incorrect filtering results	Medium	Medium	Medium	Test filtering functions
Excessive cloud resource usage	Medium	Medium	Medium	Monitor and optimize resource usage

Module 3 Final Design

Module 1 → Module 2 → Module 3

Social Media Data → Sentiment Analysis → Cloud Dashboard → Interactive Visualization → User Insights

This design makes Module 3 the presentation and user-interaction layer, connecting the Big Data Sentiment Analysis Engine with the final user.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of Module 3 – Cloud-Based Visualization Dashboard follows a modular approach. The module receives processed sentiment results from Module 2 and presents them through charts, graphs, tables, and interactive filters. The main objective is to make large-scale social media sentiment data easy to understand.

Development Steps

1. Identify dashboard requirements.
2. Collect processed sentiment results.
3. Clean and prepare data.
4. Design the dashboard interface.
5. Create sentiment charts and graphs.
6. Implement topic and time-based analysis.
7. Add interactive filters.
8. Connect the dashboard with cloud data.
9. Test visualization and filtering.
10. Deploy the dashboard.

Table 4.1 – Development Resources

Resource	Purpose
Python	Dashboard development
Pandas	Data processing
Matplotlib	Visualization
Plotly	Interactive charts
Streamlit	Dashboard development
Cloud Platform	Dashboard hosting
Cloud Storage	Data storage
Git/GitHub	Code management

Laptop	Development and testing
Internet	Cloud access

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Python is used for data processing and dashboard development. Pandas organizes and prepares sentiment data, while Matplotlib and Plotly create charts and graphs. Stream lit provides the interactive dashboard interface.

The dashboard displays positive, negative, and neutral sentiment, sentiment trends, topic-wise comparisons, keywords, and analytical tables. Interactive filters allow users to select specific dates, topics, and sentiment categories.

The dashboard can be deployed on a cloud platform to provide centralized access through a web browser.

Modern Tools Used

- Python
- Pandas
- Matplotlib
- Plotly
- Streamlit
- Cloud Platform
- Cloud Storage
- Git/GitHub

4.3 Test Plan & Verification

Testing is performed to ensure that the dashboard displays correct results and that all interactive features work properly.

Table 4.2 – Test Plan and Verification

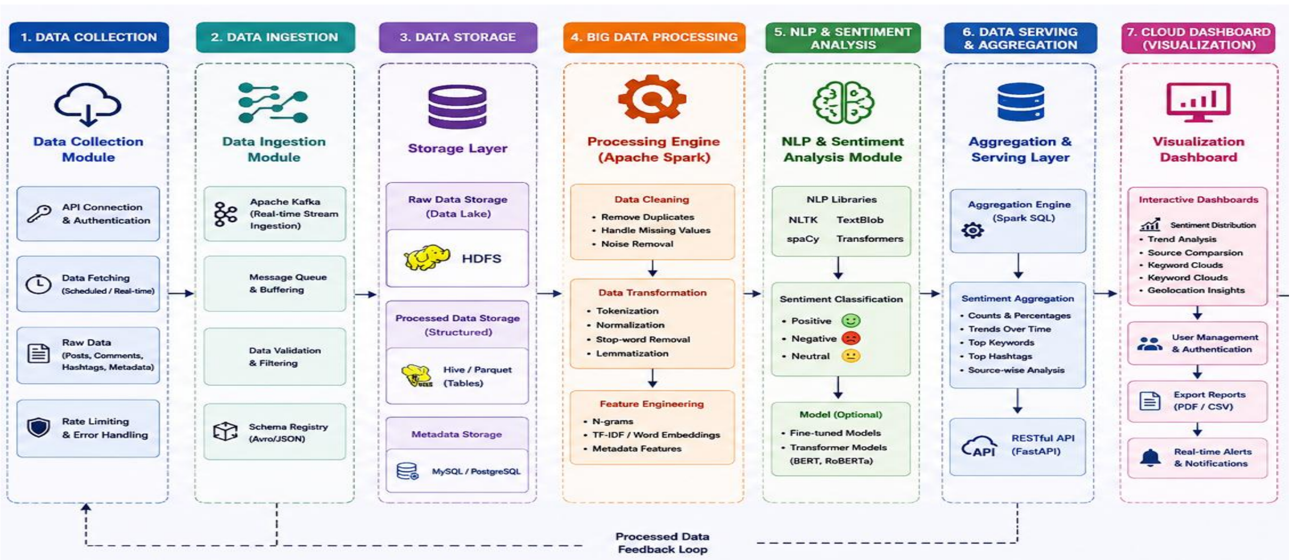
Requirement	Test Method	Acceptance Criterion
Load sentiment data	Load dataset	Data loaded successfully
Display sentiment	Open dashboard	Correct results displayed
Sentiment distribution	Check chart	Correct values displayed

Trend analysis	Select time period	Trend displayed correctly
Topic analysis	Select topic	Topic results displayed
Keyword analysis	View keywords	Keywords displayed
Date filtering	Select date	Correct data displayed
Sentiment filtering	Select category	Correct category displayed
Interactive charts	Change filters	Charts update correctly
Cloud access	Open deployed dashboard	Dashboard accessible

4.4 Results

The developed Cloud-Based Visualization Dashboard presents processed social media sentiment results in a simple and interactive format. It provides sentiment summaries, charts, graphs, tables, keywords, and trends.

Fig No:4.2 System work Flow



Dashboard Output

Visualization	**Information**
Summary Cards	Total posts and sentiment counts
Pie Chart	Sentiment distribution
Bar Chart	Topic comparison
Line Graph	Sentiment trends
Keyword Chart	Important keywords

Data Table	Detailed results
Filters	Date, topic, sentiment

4.5 Data Analysis & Engineering Judgment

The dashboard helps users understand large-scale sentiment results quickly. It provides information about positive, negative, and neutral opinions and shows how sentiment changes over time.

Interactive filters allow users to analyse specific topics and dates. Different charts are used because each visualization provides a different type of information. The dashboard is designed independently so that it can be tested before complete integration with Module 2.

4.6 Comparison with Existing Solutions & Achievement of Objectives

The proposed dashboard provides more flexibility than static reports by supporting interactive visualization and cloud-based access.

Table 4.3 – Achievement of Objectives

Objective	Evidence	Achievement
Display sentiment results	Sentiment dashboard	Fully Achieved
Display sentiment categories	Charts	Fully Achieved
Analyze trends	Line graph	Fully Achieved
Topic-wise analysis	Bar chart	Fully Achieved
Keyword analysis	Keyword chart	Achieved
Interactive filtering	Dashboard filters	Fully Achieved
Display tables	Data table	Achieved
Cloud access	Cloud deployment	Achieved
Support decision-making	Visual insights	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Simple and interactive interface.
- Clear sentiment visualization.

- Topic-wise comparison.
- Sentiment trend analysis.
- Interactive filtering.
- Cloud-based access.
- Supports faster decision-making.
- Scalable dashboard design.

Limitations

- Depends on cloud and network availability.
- Large datasets may require additional resources.
- Dashboard accuracy depends on Module 2 results.
- Real-time updates require continuous processing.

4.8 Project Planning, Roles & Budget

Project Timeline

Phase	**Activity**
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	Requirement analysis
Phase 4	Dashboard design
Phase 5	Data preparation
Phase 6	Dashboard development
Phase 7	Visualization development
Phase 8	Testing
Phase 9	Cloud deployment
Phase 10	Documentation

Team Roles

Role	**Responsibility**
Team Member 1	Dashboard and cloud setup
Team Member 2	Python/Streamlit development

Team Member 3	Data visualization
Team Member 4	Testing and documentation

Budget

Item	**Estimated Cost**
Laptop	Existing
Python	Free
Libraries	Free/Open-source
Streamlit	Free
Cloud Platform	Free tier/Educational
Internet	Existing
Total Additional Cost	Minimal

Final Flow:

Social Media Data → Sentiment Analysis → Processed Results → Cloud Dashboard → Visual Insights → Decision Making

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Cloud-Based Visualization Dashboard provides the presentation layer of the Big Data-Based Social Media Sentiment Analytics Platform. The module receives processed sentiment results from Module 2 – Sentiment Analysis & Big Data Processing Engine and converts them into meaningful charts, graphs, tables, and interactive visualizations.

The dashboard displays positive, negative, and neutral sentiment distributions, topic-wise comparisons, sentiment trends, and important keywords. Interactive filters allow users to explore the results based on date, topic, and sentiment category.

The proposed module makes large-scale social media analytics easier to understand and supports faster data-driven decision-making. The cloud-based approach provides centralized access to analytical results and enables future scalability.

5.2 Future Work

Module 3 can be improved in future versions by:

- Adding real-time social media data updates.
- Providing advanced interactive visualizations.
- Adding more filtering and drill-down options.
- Supporting additional sentiment categories.
- Integrating predictive sentiment trend analysis.
- Adding automated alerts for significant sentiment changes.
- Improving mobile and responsive dashboard design.
- Deploying the dashboard on scalable cloud infrastructure.
- Supporting larger social media datasets.
- Adding multilingual sentiment visualization.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned how to design and develop a cloud-based dashboard.
- Learned how to convert analytical data into meaningful visual information.
- Understood different types of charts and graphs and their applications.
- Learned interactive filtering techniques for data exploration.
- Understood the connection between sentiment analysis and data visualization.
- Learned the importance of presenting Big Data in a simple and understandable format.
- Engineering & Professional Skills Developed
- Dashboard design
- Python programming
- Data visualization
- Data analysis
- Cloud-based application development
- Testing and debugging
- Problem-solving
- Technical documentation
- Team collaboration

Individual Reflection

During the development of Module 3, the main engineering challenge was presenting large-scale sentiment analysis results in a simple and understandable manner. Selecting suitable charts and designing effective filters helped improve the usability and clarity of the dashboard.

The modular architecture made it possible to develop and test the visualization component independently before integrating it with Module 2. This approach improved the development process and helped ensure that the dashboard could effectively present the processed analytical results.

Through this module, practical knowledge of Python, data visualization, interactive dashboards, and cloud-based applications was improved. The project also provided valuable experience in analysing data from different perspectives and presenting meaningful insights to users.

If the system were redesigned in the future, greater emphasis would be placed on real-time data.

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APPENDICES

APPENDIX A – DETAILED CALCULATIONS

For Module 3 – Cloud-Based Visualization Dashboard, the main calculations are related to sentiment distribution, sentiment percentages, and topic-wise analysis.

A.1 Sentiment Percentage Calculation

The percentage of a particular sentiment can be calculated as:

Sentiment Percentage = (Number of posts with particular sentiment / Total number of posts) × 100

Example:

Total posts = 1,000
Positive posts = 600

Positive Percentage = (600 / 1000) × 100 = 60%

Similarly, negative and neutral sentiment percentages can be calculated.

A.2 Sentiment Score

A sentiment score can be used to represent the strength or direction of sentiment. The processed sentiment scores received from Module 2 are used by the dashboard for visualization and comparison.

APPENDIX B – SYSTEM WORKFLOW

USER

↓

MODULE 1

Social Media Data Collection

↓

MODULE 2

Sentiment Analysis &

Big Data Processing Engine

↓

Processed Sentiment Data

↓

MODULE 3

Cloud-Based Visualization

Dashboard

```
          ↓
┌──────────────────────────────┐
│ Charts / Graphs / Tables     │
│ Filters / Sentiment Trends   │
└──────────────┬───────────────┘
          ↓
┌──────────────────────────────┐
│ User Insights & Decision     │
│ Making                       │
└──────────────────────────────┘
```

APPENDIX C – SAMPLE DASHBOARD CODE

```
import streamlit as st
import pandas as pd
import plotly.express as px
st.title("Big Data-Based Social Media Sentiment Analytics")
st.subheader("Cloud-Based Visualization Dashboard")
data = pd.DataFrame({
    "Sentiment": ["Positive", "Negative", "Neutral"],
    "Count": [600, 250, 150]
})
```

```python
st.write("Sentiment Summary")

st.dataframe(data)

fig = px.pie(

    data,

    names="Sentiment",

    values="Count",

    title="Sentiment Distribution"

)

st.plotly_chart(fig)

selected = st.selectbox(

    "Select Sentiment",

    ["All", "Positive", "Negative", "Neutral"]

)

if selected == "All":

    st.write(data)

else:

    st.write(data[data["Sentiment"] == selected])
```

APPENDIX D – DATA SPECIFICATION

Sample Sentiment Data

S. No.	Topic	Sentiment	Score
1	Product	Positive	0.91
2	Service	Negative	-0.82
3	Price	Neutral	0.05
4	Delivery	Positive	0.87
5	Quality	Negative	-0.74
6	Support	Positive	0.79

Data Fields

Parameter	Data Type	Purpose
Topic	String	Identifies discussion topic
Sentiment	String	Positive/Negative/Neutral
Score	Float	Represents sentiment strength
Date	Date	Used for trend analysis
Keywords	String	Identifies important terms

APPENDIX E – TEST DATA

Test ID	Topic	Sentiment	Filter	Result
T01	Product	Positive	Positive	Displayed
T02	Service	Negative	Negative	Displayed
T03	Price	Neutral	Neutral	Displayed
T04	Delivery	Positive	All	Displayed
T05	Quality	Negative	Topic	Displayed
T06	Support	Positive	Date	Displayed

Purpose of Test Data

The test data is used to verify:

- Correct loading of sentiment data.
- Correct sentiment classification display.
- Accurate charts and graphs.
- Proper filtering.
- Topic-wise analysis.
- Date-based analysis.
- Correct dashboard output.

APPENDIX F – ETHICAL CONSIDERATIONS

The project processes social media data, so responsible data handling is important.

The following principles should be followed:

1. Social media data should be collected only for legitimate project purposes.
2. Personal information should not be unnecessarily stored.
3. User privacy should be protected.
4. Sensitive information should not be exposed through the dashboard.
5. Data should be processed according to applicable policies and permissions.
6. Dashboard access should be restricted to authorized users.
7. The system should avoid making misleading conclusions from sentiment results.

Institutional Approval

If institutional approval is required, the relevant approval details should be included.

If approval is not applicable:

"Institutional approval was not applicable to this software-based academic project because the system analyses social media data and does not involve clinical intervention or direct physical experimentation."

APPENDIX G – PROJECT PROGRESS RECORD

Meeting	Discussion	Work Completed	Next Activity
1	Project topic selection	Topic finalized	Literature review
2	Project requirements	Requirements identified	Module design
3	Module 3 discussion	Dashboard requirements identified	Dashboard design
4	Python implementation	Initial dashboard developed	Visualization
5	Visualization	Charts and graphs created	Interactive filters
6	Testing	Test cases completed	Error correction
7	Module integration	Module 2 connected with Module 3	Final testing
8	Report preparation	Documentation completed	Final submission

APPENDIX H – FEEDBACK FORM

S.No.	Evaluation Area	Rating / Feedback
1	Dashboard usability	__________
2	Visualization clarity	__________
3	Ease of filtering	__________
4	Sentiment understanding	__________
5	Dashboard performance	__________
6	Overall usability	__________
7	Suggestions	__________

Sample Feedback Questions

1. Is the dashboard easy to understand?
2. Are sentiment results clearly displayed?
3. Are the charts and graphs useful?
4. Are the filters easy to use?
5. Is the dashboard responsive?
6. What additional features should be included?

APPENDIX I – INDIVIDUAL CONTRIBUTION EVIDENCE

Student	Responsibility	Development	Testing	Report Contribution
Student 1	Module 1 – Data Collection	Data collection	Module testing	Chapter 1
Student 2	Module 2 – Sentiment Analysis	Processing and analysis	Model testing	Chapter 2
Student 3	Module 3 – Visualization Dashboard	Dashboard development	Dashboard testing	Chapter 3, 4
Student 4	Integration & Documentation	Module integration	System testing	Chapter 5, References
Total				100%

Module 3 Individual Contribution

If you are responsible for Module 3 – Cloud-Based Visualization Dashboard, your contribution can be described as:

- Identified dashboard requirements.
- Designed the dashboard interface.
- Implemented data visualization.
- Created charts and graphs.
- Implemented interactive filters.
- Integrated processed sentiment data from Module 2.
- Tested dashboard functionality.
- Analysed visualization results.

- Prepared documentation related to Module 3.

Outcome

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)